NEXUS

Causal Intelligence Platform — Risk Report

Repository: https://github.com/postgres/postgres

Analysis Window: 2 years

Generated: 2026-05-16T22:49:43

Total DPRs: 28

CTG Edges: 25

Decay Alerts: 5

Executive Summary

This report analyzes 15 Decision Provenance Records (DPRs) across the PostgreSQL codebase. **5 DPRs** have critical risk scores (>=70), **4 assumptions** are actively decaying, and **5 items** require immediate remediation (P0).

Risk Scores — Top 10

DPR	Title	Component	Risk	Blast	Decay	Active?
DPR-003	MVCC via Tuple Versioning in H	MVCC	93	100.0	100.0	YES
DPR-004	32-bit Transaction ID with Wra	MVCC	93	100.0	100.0	YES
DPR-006	Vacuum as Separate Maintenance	Storage	93	100.0	100.0	YES
DPR-008	Autovacuum as Background Daemo	Autovacuum	87	100.0	100.0	YES
DPR-001	8KB Fixed Page Size	Storage	72	100.0	50.0	no
DPR-005	HOT (Heap-Only Tuple) Updates	MVCC	67	75.0	50.0	no
DPR-009	Multi-Process Architecture	ProcessModel	62	100.0	50.0	no
DPR-011	Shared Buffer Pool with Clock-	Storage	62	100.0	50.0	no
DPR-007	Write-Ahead Logging (WAL) for	WAL	60	100.0	50.0	no
DPR-010	Snapshot Isolation as Default	MVCC	58	75.0	25.0	no

Active Decay Alerts

[DECAYING] **DPR-003**: Vacuum can keep up with tuple version creation rate

Evidence: High-update workloads on cloud storage creating bloat faster than vacuum can clean

[DECAYING] **DPR-004**: Databases won't exceed 2B transactions between vacuum freeze operations

Evidence: High-throughput OLTP systems (100K+ TPS) can generate 2B transactions in ~6 hours

[DECAYING] **DPR-006**: Tables are small enough to vacuum in reasonable time

Evidence: TB-scale tables taking 6-12 hours to vacuum, cloud storage IOPS limits making vacuum expensive

[DECAYING] **DPR-008**: Default thresholds (20% of table) are appropriate for most workloads

Evidence: 20% of 1TB table = 200GB of changes before vacuum triggers, causing massive bloat

[MONITORING] **DPR-011**: shared_buffers should be 25-40% of system RAM

Evidence: Cloud instances with limited RAM make large buffer pools expensive, NVMe reducing cache miss penalty

Remediation Backlog

[P0] DPR-003: MVCC via Tuple Versioning in Heap — Risk: 93

- Address 1 active decay alert(s)
- Review 5 workarounds for consolidation
- Add blast radius monitoring and alerts
- Schedule quarterly assumption review

[P0] DPR-004: 32-bit Transaction ID with Wraparound — Risk: 93

- Address 1 active decay alert(s)
- Review 5 workarounds for consolidation
- Add blast radius monitoring and alerts
- Schedule quarterly assumption review

[P0] DPR-006: Vacuum as Separate Maintenance Process — Risk: 93

- Address 1 active decay alert(s)
- Review 5 workarounds for consolidation
- Add blast radius monitoring and alerts
- Schedule quarterly assumption review

[P0] DPR-008: Autovacuum as Background Daemon — Risk: 87

- Address 1 active decay alert(s)
- Review 5 workarounds for consolidation
- Add blast radius monitoring and alerts
- Schedule quarterly assumption review

[P0] DPR-001: 8KB Fixed Page Size — Risk: 72

- Review 3 workarounds for consolidation
- Add blast radius monitoring and alerts

[P1] DPR-005: HOT (Heap-Only Tuple) Updates — Risk: 67

- Review 4 workarounds for consolidation

[P1] DPR-009: Multi-Process Architecture — Risk: 62

- Review 4 workarounds for consolidation
- Add blast radius monitoring and alerts

[P1] DPR-011: Shared Buffer Pool with Clock-Sweep — Risk: 62

- Review 4 workarounds for consolidation
- Add blast radius monitoring and alerts

[P1] DPR-007: Write-Ahead Logging (WAL) for Durability — Risk: 60

- Review 4 workarounds for consolidation
- Add blast radius monitoring and alerts

[P1] DPR-010: Snapshot Isolation as Default — Risk: 58

- Review 3 workarounds for consolidation

[P1] DPR-012: Checkpoint-Based Crash Recovery — Risk: 57

- Review 4 workarounds for consolidation

[P2] DPR-002: TOAST - The Oversized-Attribute Storage Technique — Risk: 52

- Review 3 workarounds for consolidation

[P2] DPR-013: Full-Page Writes for Torn Page Protection — Risk: 51

- Review 4 workarounds for consolidation

[P2] DPR-014: Streaming Replication via WAL Shipping — Risk: 43

- Review 4 workarounds for consolidation

[P2] **DPR-015**: External Connection Pooling Required — Risk: 43

- Review 4 workarounds for consolidation

Knowledge Concentration — SPOF Alerts

Original PostgreSQL team → DPR-001: 8KB Fixed Page Size (100.0% ownership, critical blast radius)

Engineer 'Original PostgreSQL team' holds 100.0% of causal knowledge about 8KB Fixed Page Size (Storage). Their departure is a critical SPOF.

Jan Wieck → DPR-002: TOAST - The Oversized-Attribute Storage Technique (66.7% ownership, high blast radius)

Engineer 'Jan Wieck' holds 66.7% of causal knowledge about TOAST - The Oversized-Attribute Storage Technique (Storage). Knowledge transfer recommended.

Tom Lane → DPR-004: 32-bit Transaction ID with Wraparound (66.7% ownership, critical blast radius)

Engineer 'Tom Lane' holds 66.7% of causal knowledge about 32-bit Transaction ID with Wraparound (MVCC). Knowledge transfer recommended.

Tom Lane → DPR-006: Vacuum as Separate Maintenance Process (66.7% ownership, critical blast radius)

Engineer 'Tom Lane' holds 66.7% of causal knowledge about Vacuum as Separate Maintenance Process (Storage). Knowledge transfer recommended.

Alvaro Herrera → DPR-008: Autovacuum as Background Daemon (66.7% ownership, critical blast radius)

Engineer 'Alvaro Herrera' holds 66.7% of causal knowledge about Autovacuum as Background Daemon (Autovacuum). Knowledge transfer recommended.

Original PostgreSQL team → DPR-009: Multi-Process Architecture (100.0% ownership, critical blast radius)

Engineer 'Original PostgreSQL team' holds 100.0% of causal knowledge about Multi-Process Architecture (ProcessModel). Their departure is a critical SPOF.

Vadim Mikheev → DPR-010: Snapshot Isolation as Default (100.0% ownership, high blast radius)

Engineer 'Vadim Mikheev' holds 100.0% of causal knowledge about Snapshot Isolation as Default (MVCC). Their departure is a critical SPOF.

Tom Lane → DPR-011: Shared Buffer Pool with Clock-Sweep (66.7% ownership, critical blast radius)

Engineer 'Tom Lane' holds 66.7% of causal knowledge about Shared Buffer Pool with Clock-Sweep (Storage). Knowledge transfer recommended.

Tom Lane → DPR-012: Checkpoint-Based Crash Recovery (66.7% ownership, high blast radius)

Engineer 'Tom Lane' holds 66.7% of causal knowledge about Checkpoint-Based Crash Recovery (WAL). Knowledge transfer recommended.

Vadim Mikheev → DPR-013: Full-Page Writes for Torn Page Protection (66.7% ownership, high blast radius)

Engineer 'Vadim Mikheev' holds 66.7% of causal knowledge about Full-Page Writes for Torn Page Protection (WAL). Knowledge transfer recommended.

Heikki Linnakangas → DPR-014: Streaming Replication via WAL Shipping (66.7% ownership, high blast radius)

Engineer 'Heikki Linnakangas' holds 66.7% of causal knowledge about Streaming Replication via WAL Shipping (Replication). Knowledge transfer recommended.

pgBouncer: Marko Kreen → DPR-015: External Connection Pooling Required (66.7% ownership, high blast radius)

Engineer 'pgBouncer: Marko Kreen' holds 66.7% of causal knowledge about External Connection Pooling Required (ProcessModel). Knowledge transfer recommended.

Counterfactual Simulation Summary

[EXTREME] What if PostgreSQL used 16KB pages instead of 8KB?

Affected: 10 DPRs across 4 components

[EXTREME] What if PostgreSQL used undo logs instead of in-heap tuple versioning?

Affected: 5 DPRs across 3 components

[EXTREME] What if PostgreSQL used 64-bit transaction IDs?

Affected: 5 DPRs across 3 components

[HIGH] What if PostgreSQL adopted multi-threaded architecture?

Affected: 4 DPRs across 3 components

[MEDIUM] What if PostgreSQL used shadow paging instead of WAL?

Affected: 3 DPRs across 2 components